Bhavani Prasad Kola

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PROFESSIONAL SUMMARY

Computer Science and Engineering graduate with hands-on experience in MERN-stack web development and AI/ML projects. Skilled in building React interfaces, integrating REST APIs, developing Node.js/Express services, and working with MongoDB. Project experience includes AI voice-agent interfaces, real-time sign-language recognition, and computer-vision-based insect detection. Brings a foundation in Java, data structures and algorithms, and object-oriented programming.

EDUCATION

Lakireddy Bali Reddy College of Engineering **2022 – 2026**
B.Tech, Computer Science and Engineering | CGPA: 8.37/10 (79.52%) *Mylavaram, Andhra Pradesh*

KBN Junior College **2019 – 2021**
Intermediate, Mathematics, Physics and Chemistry | 92% *Vijayawada, Andhra Pradesh*

TECHNICAL SKILLS

Programming: Java, JavaScript, Python

Frontend: React.js, HTML5, CSS3

Backend & APIs: Node.js, Express.js, REST APIs

Databases: MongoDB, SQL

AI/ML : Machine Learning, deep learning

Tools: Git

Computer Science: Data Structures and Algorithms

PROJECTS

Real-Time Sign Language Recognition Using Deep Learning **2025–2026**
Technologies: Python, TensorFlow, Keras, MediaPipe, MobileNetV2, CBAM, OpenCV

- Developed a computer-vision pipeline using OpenCV and MediaPipe to capture webcam input, detect hand landmarks, and extract gesture features.
- Built a TensorFlow/Keras deep-learning workflow for gesture classification, including preprocessing and confidence-based predictions.
- Added gesture-to-text and optional text-to-speech output to support an end-to-end sign-recognition interaction.

API Key Management System **2026**
Technologies: React.js, Node.js, Express.js, MongoDB, JWT

- Developed a full-stack dashboard for organizing and managing API credentials through a React.js interface and Node.js/Express backend.
- Implemented role-based access control, encrypted key storage, and audit logging to support controlled access and credential security.
- Integrated MongoDB to persist application data and connect frontend workflows with backend services.

English AI Voice Call Agent **2026**
Technologies: React.js, Node.js, Express.js, REST APIs, GPT-4.1 Mini, Retell AI

- Built a responsive React.js interface for configuring voice agents, controlling calls, viewing live conversation details, and reviewing call history.
- Integrated REST APIs with backend services to manage agent settings, call sessions, and conversation records.
- Integrated GPT-4.1 Mini with Retell AI to support context-aware voice conversations using speech recognition and text-to-speech capabilities.

AI-Powered Smart Drone for Precision Agriculture **2025**
Technologies: Python, Computer Vision, MobileNetV2, Embedded Systems, IoT

- Developed an insect-detection pipeline using a dataset of 50,000+ insect images.
- Designed a precision-spraying approach to target affected crop zones and reduce unnecessary pesticide application.
- Co-invented the AI-based precision-spraying system, which is described in the supplied resume as a published/registered patent.

INTERNSHIPS / EXPERIENCE

Biztron Softech Ltd. **Jun 2024 – Jul 2024**
MERN Stack Intern *Vijayawada, Andhra Pradesh*

- Developed reusable React.js components to support consistent and responsive web interfaces.
- Integrated RESTful APIs to connect frontend components with backend services.
- Gained hands-on experience across the MERN stack, contributing to frontend development and API integration.

SmartBridge (APSCHE) **Dec 2024 – May 2025**
AI & ML Virtual Intern *Remote*

- Applied supervised and unsupervised learning techniques, including regression, classification, clustering, and dimensionality reduction, to datasets.
- Worked through ML workflow stages including data acquisition, preprocessing, feature engineering, model training, hyperparameter tuning, and evaluation.
- Completed an end-to-end machine-learning project involving model deployment workflows.

SmartBridge **May 2025 – Jul 2025**

Salesforce Developer Intern *Remote*

- Developed Lightning Web Components (LWC) and Apex classes for Salesforce business logic.
- Automated business workflows using Salesforce Flow, Process Builder, and custom triggers.

CERTIFICATIONS

- AWS Certified Cloud Practitioner – Amazon Web Services
- Salesforce Developer Certification – SmartBridge
- ServiceNow Certified System Administrator (CSA) – ServiceNow
- Smart Coder Certification – Smart Interviews (DSA and Problem Solving)

ACHIEVEMENTS

- **1st Prize**, National-Level 48-Hour AI Hackathon, VR Siddhartha College of Engineering, for an insect-detection and AI-powered pesticide-spraying drone system.
- **Patent:** Co-inventor on an AI-based precision pesticide-spraying system. Confirm the exact patent publication/registration status and number before submitting.
- **Research Presentation:** Paper accepted and presented at the ICICT International Conference; the supplied resume states it was included in the proceedings.